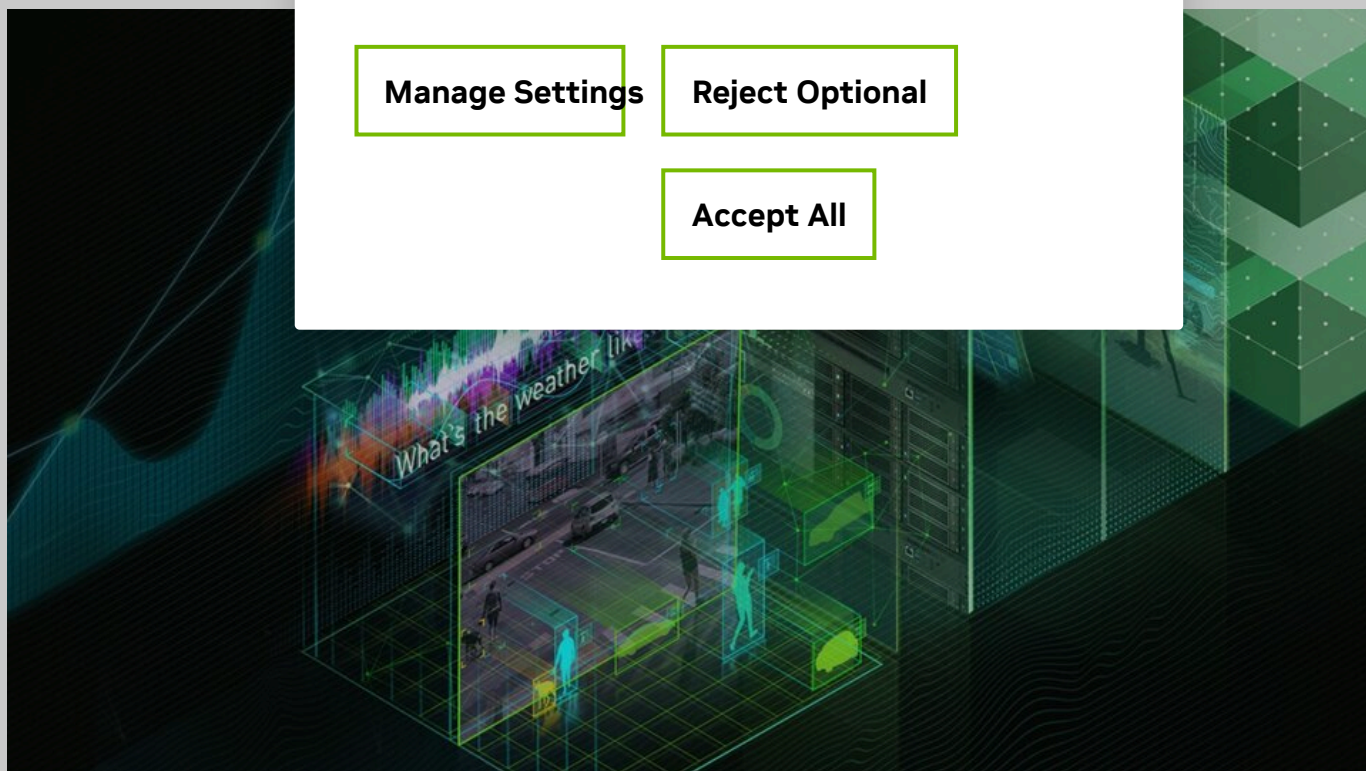


Agentic AI / Generative AI

Introducing Reusability NVIDIA

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Jan 16, 2025

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AI-Generated Summary



historical context in LLM serving for generation of the next set of tokens. Caching these key and value elements from previous tokens avoids expensive recomputation and

effectively leads to a significant reduction in memory usage, especially with

the size of the language models and the length of the input and output sequences.

lengths, leading to a significant

NVIDIA TensorRT-LLM

balance between generating

LLM is an open-source

numerous popular LLMs

caching includes several

cache, circular buffer

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In this post, we dive

TensorRT-LLM. These

provide visibility into

aware routing.

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che, and

like KV cache

Priority-based KV cache eviction

When an LLM request has completed, the KV cache blocks associated with these requests are stored. Given the bounded size of the KV cache, some cached blocks may need to be evicted to make room for new sequences. By default, eviction follows a least recently used (LRU) policy.

Priority-based eviction is a new feature of the TensorRT-LLM Executor API that enables users to influence how blocks are selected for eviction. Users can specify two attributes that guide block eviction: priority and duration. The priority value sets the relative retention priority (how important it is to retain that block in the cache), and the duration value sets how long this priority level should apply for.

```
struct TokenRangeRetentionConfig {  
    # The beginning of this range  
    start: int
```

```
priority: int
# The duration this priority should apply for
duration: optional<int>

}

# Optional parameters
struct KvCacheRet
{
    # List of priority ranges: list<int>
    # Priority as decode_priority
    # Duration the decode_duration
}
```

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The priority based-workload to improve. For example, the decode cache as long as possible should persist with

at their
be reused.
to stay in the
request

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Figure 1. Leverage knowledge of your workloads to better control the KV cache reuse opportunity with the priority-based eviction API in NVIDIA TensorRT-LLM

For each request, you can specify a priority and duration value for discrete ranges of tokens in the input context, along with a priority and duration for blocks allocated during the decode phase. The priority level of a range of tokens applies until the duration has passed after no period of reuse, or until the blocks corresponding to these ranges have been evicted.

When choosing blocks to be evicted, TensorRT-LLM considers the priority levels of tokens within the block. For example, a request with a 500- token system prompt can set the token range $[0, 500)$ to the maximum priority. This way, the cache blocks corresponding to these tokens will only be evicted if absolutely necessary. Alternatively, if you know that blocks will never be reused, you can set the blocks of this request to the lowest priority to ensure that they are evicted first, before other blocks.

benchmarks show priority-based eviction increasing cache hit rate by around 20% and varies based on the workload.

```
# Priority-based
```

```
#Example 1: One-o
```

```
KvCacheRetentionC
[TokenRangeRe
decode_priori
)
```

```
#Example 2: High
```

```
KvCacheRetentionC
[TokenRangeRe
)
```

```
#Example 3: Retai
```

```
KvCacheRetentionC
[TokenRangeRe
decode_priori
```

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```
seconds
```

```
9s)],
```

KV cache event API

In large-scale LLM-powered applications, deployers often provision multiple serving instances of a model to distribute incoming requests. This raises the question, which instance should process new requests? Requests are often routed to balance load to ensure efficient utilization and quick processing of any request. The size of the KV cache on any instance represents the capacity to grow and accept new work.

However, load-based routing may not be optimal. If a moderately loaded instance has already computed and cached the keys and values for a new request, routing the request to this instance might still be preferred to optimize for cache reuse. The KV cache event API enables request routing systems to track which instances have cached or evicted blocks, enabling more intelligent reuse and greater performance.

The TensorRT-LLM [Executor API](#) now exposes a means of tracking updates to the KV cache.

```

}

struct StoredBlock {
    blockHash: id
    tokens: list<
    loraId: id
    cacheLevel: id
    priority: int
}

struct StoredData {
    parentHash: id
    blocks: list<
}

struct RemovedData {
    blockHashes:
}

# Set the max size of the cache
kv_cache_config =

executor_config =

executor = Executor

# Get an event manager
eventManager = executor.getKvCacheEventManager()

# Wait for new events. Once it returns, it implicitly clears the internal queue of events
events = eventManager.getLatestEvents()

```

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When a cache block is stored for reuse, removed, or updated, an event is emitted. These events can be consumed in real time by an application to get an eventually consistent view of the current state of the TensorRT-LLM KV cache. This is especially useful for tracking KV cache reuse opportunities. It can be used on the scale of a single executor to anticipate which requests will have more reuse, or aggregated across many executors to make KV-aware routing and scheduling decisions (Figure 2).

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Figure 2. Optimize the KV cache reuse opportunity with events-driven KV-aware routing of requests using the KV cache event API in NVIDIA TensorRT-LLM

With the introduction of priority-based eviction and event-aware routing for KV cache reuse and management, TensorRT-LLM provides you with levers that enable fine-grained control of the KV cache so you can use the knowledge of your workloads to optimize KV cache management.

NVIDIA TensorRT-LLM provides several optimizations to efficiently deploy your generative AI applications across edge devices, data center, and workstation. It leverages techniques like model quantization, kernel cache reuse on the GPU, and workload distribution to optimize inference on the same workload, reducing latency and improving throughput.

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John Thomson is an intern on the Deep Learning Algorithms team at NVIDIA. He's currently in his third year of Computer Engineering at the University of Waterloo. His area of focus is optimizing LLM inference on structured workloads.

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About Anjali Shah

Anjali Shah is a senior deep learning scientist at NVIDIA within the Developer Advocate Engineering group helping clients build generative AI solutions. Early in her career, as a software engineer, she built mission-critical platforms for the world's leading financial services firms. She then spent several years in the healthcare sector, architecting and implementing large scale healthcare (EHR) systems. Before joining NVIDIA, she spent several years at a leading tech company, working across different industries helping clients build innovative data and AI solutions. She has a Ph.D. in biomedical

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Notable



far

Sharing an inspiration, priority-based KV cache allows a user to retain a specific set of tokens. Will this be especially useful for agent workload, since tokens such as “planning the next steps”, “checking the results”, etc., will be repeatedly called?

Continue the discussion at
forums.developer.nvidia.com

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